Evaluation and Program Planning 98 (2023) 102268

Contents lists available at ScienceDirect

Evaluation and Program Planning

journal homepage: www.elsevier.com/locate/evalprogplan

Evaluating and sustaining Coordinated Specialty Care for a recent onset of psychosis in non-academic-affiliated community mental healthcare settings

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A R T I C L E I N F O

Keywords:
Coordinated Specialty Care
First-episode psychosis
Program evaluation
Service participation
Community mental health

A B S T R A C T

Purpose: To improve sustainability of Coordinated Specialty Care (CSC) for a recent onset of psychosis, a better understanding is needed regarding how non-academic-affiliated community mental health centers blend CSC service elements and select key performance metrics to evaluate their approach.
Methods: A quality and evaluation team embedded within a large community mental health center partnered with CSC site leadership to implement CSC and design a program evaluation strategy informed by CSC research literature. Clinical, family, vocational, and psychiatry services participation, exits, key performance indicators, and standardized measures were examined for participants (n = 47) enrolled for 12-months.
Results: Mean service participation was 55 h (SD = 23.5) in the first 12-months (approximately 4.70 h/month). All participated in clinical; 87% in psychiatry; 67% in vocational; and 57% in family services. Sixty-one percent had planned service exits; 39% had unplanned exits. Across the 12-months, 83% were employed or in school; 72% were not psychiatric hospitalized.
Conclusions: CSC participation and outcomes were similar to the limited research examining both together. Understanding service participation and provider adjustments to sustain CSC is critical in community mental healthcare settings that rely on fee-for-service billing mechanisms. Findings have implications for national CSC data harmonization and sustainability efforts.

It is incredibly exciting when successful community mental health intervention research trials drive policy change that produces widespread implementation of evidence-based services. This is what happened with the National Institute of Health *Recovery After an Initial Schizophrenia Episode* (RAISE) early psychosis trials in the United States over the last decade. RAISE examined the impact of a multidisciplinary team-based care coordination model for those with a recent onset of psychosis (Heinssen et al., 2014). Two large multi-site studies were conducted (Kane et al., 2016; Dixon et al., 2015) and findings translated into federal policy. The Substance Abuse and Mental Health Services Administration (SAMHSA) mandated a percentage of each state's federal mental health block grant fund first-episode psychosis care (NRI, 2018). Today, Coordinated Specialty Care (CSC) for a recent onset of psychosis is available in most states (NAMI, 2017). However, translating science to sustained practice in community mental health in the U.S. comes with significant barriers. Community mental health systems and providers are often under-resourced in regards to funding, staff, and technology. Most community mental health agencies rely on Medicaid funding for the majority of their operating costs (Dybdal et al., 2014). To date, Medicaid is typically a fee-for-service (as opposed to pay-for-performance) billing mechanism. The RAISE studies and other CSC research have captured positive outcomes in mental health and quality of life (e.g., Kane et al., 2016; Dixon et al., 2015), but few have published details around CSC service participation. Community mental health agencies (and state systems) need to understand service participation in fee-for-service billing environments to sustain CSC

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https://doi.org/10.1016/j.evalprogplan.2023.102268
Received 4 October 2021; Received in revised form 19 December 2022; Accepted 24 February 2023
Available online 28 February 2023

implementation. There is also a need to better understand how community mental health agencies engage in CSC evaluation and quality improvement in non-academic affiliated settings.

What is CSC? Coordinated Specialty Care (CSC) is an evidence-informed approach to treating a recent onset of psychosis. CSC was developed to engage individuals (and their families) as quickly as possible following an initial psychotic episode (Dixon et al., 2018) to prevent long-term struggles associated with schizophrenia (e.g., psychiatric hospitalizations, unemployment, poverty, homelessness, disability benefit reliance; Charlson et al., 2018; Salomon et al., 2015). CSC is a team-based approach that leverages intensive early intervention, shared-decision making and evidence-based practice blending to concurrently address mental health symptoms and life functioning among individuals experiencing a recent onset of psychosis (Dixon et al., 2018; Goldman, 2020; Heinssen et al., 2014). CSC roles tend to include: a clinically licensed team leader, a clinically-licensed therapist, an Employment and Education Specialist, and a prescriber (either a psychiatrist or nurse practitioner) (Addington, 2021; Bello et al., 2017; Mueser et al., 2015). However, some CSC Models also include: Case Managers (i.e., NAVIGATE), Peer Support Specialists (i.e., OnTrackNY) and Occupational Therapists (i.e., PIER) (NASMHPD, 2017). Evidence-informed & -based practices used in CSC include: individual and family psychoeducation, Cognitive Behavior Therapy for psychosis, case management, 24-hour on-call crisis support, and Supported Employment and Supported Education (Addington, 2021). CSC also has a community outreach and education component that requires strategic education and partnership with hospitals, primary care and mental health providers, schools, and local organizations that serve youth to identify individuals with first-episode psychosis and swiftly connect them to CSC.

Federally-funded research finds CSC to be beneficial for symptom reduction, quality of life, and cost-savings. In a comparison CSC study (n = 223) versus non-CSC participants (n = 181), clinician-rated measures over-time showed significant decreases in mental health symptom severity and increases in quality of life, in particular for those with shorter durations of untreated psychosis (Kane et al., 2015). Within the first year of CSC enrollment (n = 235), psychiatric hospitalizations decreased by 60%; and education and/or employment engagement increased by 40% (Nossel et al., 2018). Treatment costs were lower among those with FEP participating in CSC compared to usual treatment (Murphy et al., 2018; Rosenheck et al., 2016). States with strong university partnerships also find CSC promising. Both Washington (n = 112) and Pennsylvania (n = 230) documented decreased symptoms and psychiatric hospitalizations, and increased vocational engagement and quality of life among CSC participants (Oluwoye et al., 2020; Westfall et al., 2020).

In 2015, unprecedented U.S. policy mandated states to use 5% (expanded to 10% in 2016 and continued annually since) of their mental health block grant to fund early intervention psychosis care (NRI, 2018). In addition to the national CSC implementation surge, concurrent national learning communities, trainings, and CSC implementation guidance also emerged (EPINET, 2021; NASMHPD, 2021; NRI, 2019). These resources included multiple CSC Models, but most prominently featured NAVIGATE, OnTrackNY, EASA, EDAPT and PIER Models. States and agencies had the opportunity to shape their on-the-ground CSC programs in real-time. Some states fully adopted and enforced the implementation of one CSC model, some mixed elements of various CSC Models that aligned with their state's mental health policy and practice philosophies (and billing mechanisms), and some states provided little to no CSC oversight to providers beyond funding (NASMHPD, 2017; 2018).

Little has been published on publicly-based, non-academic CSC model designs, nor how these fully-community based CSC sites evaluate their efforts – or make adjustments to improve their program performance. Key Performance Indicators (KPIs) are typically driven by stakeholders and funders (Lauriks et al., 2012), and although they may align with or be informed by published research, they warrant examination if CSC is truly to be sustained across the U.S. To date, service participation rates matter just as much (if not more than outcomes) in most community mental health settings because Medicaid and commercial insurance pay for services delivered, not outcomes achieved (Dybdal et al., 2014). Further, in multidisciplinary service models, there are multiple service providers working together to engage participants and support their goal attainment. In CSC, this includes: psychiatry/prescribing, clinical/therapy, case management/care coordination, and employment and education service lines (Addington, 2021), which all are typically billed at separate Medicaid and commercial insurance rates depending on state and insurance company. Examining participation in the different CSC service types over-time is critical for financially sustaining CSC, both for covering CSC in the current fiscal environment and for advocating for policy changes to improve coverage. Successful implementation of evidenced-based interventions in the community depends largely on contextual factors (Bauer & Kirchner, 2020). The 12–52% CSC premature disengagement rate (Mascayano et al., 2021) is troubling. A better understanding of both participation and outcomes over-time is necessary for CSC providers to effectively alter their approaches to prevent disengagement. There is much to be learned about how real-world providers translate science into sustainable practice through blending CSC model elements, evaluating and refining their approaches.

There have been various efforts to evaluate CSC implementation and impact. Most efforts are at the national, state or county mental health authority level (NASMHPD, 2018), in part related to state/county department contracting with community providers to deliver CSC. Some states have partnered with university researchers (e.g., Texas, Pennsylvania, California, New York, Washington; NASMHPD, 2018) that have led to published state-level program evaluation findings in the scientific literature (Niendam et al., 2019; Nossel et al., 2018; Oluwoye et al., 2019). Although external evaluation efforts led by an academic research team are valuable in understanding policy impact on a target population, there are few deep dives into the internal program evaluation efforts of agencies in implementing CSC. One study does investigate CSC program design, service participation and outcomes at one site, but the CSC site is embedded within a university healthcare setting, and the evaluation is led by an academic research team (North et al., 2019). It is critical that local community provider-led program evaluation efforts are investigated beyond CSC sites that have resources, technology and expertise of a university setting.

1. CSC model development and implementation in real world setting

Thresholds, the largest community mental health center (CMHC) in Illinois, designed and implemented two CSC "MindStrong" teams: one in Cook County (where Chicago is located) and one in DuPage County (Chicago's western suburbs). MindStrong is funded through Medicaid, commercial insurance and Illinois's mental health block grant. NAVIGATE, OnTrackNY, PIER, and EASA CSC approaches (NASMHPD, 2017) informed MindStrong design. Thresholds participated in national CSC trainings and learning communities (National Council for Behavioral Health, 2018; NASMHPD, 2017), met with CSC developers and evaluators, and visited CSC sites in several states. MindStrong also adheres to Illinois FIRST.IL CSC requirements introduced in 2018 (IL DHS, 2020).

When MindStrong started, each site had: 1.0 full-time effort (FTE) licensed masters-level Team Leader, 2.0 FTE licensed masters-level Clinicians, 0.50 FTE Employment & Education Specialist, and 0.20 Prescriber. Both teams share a 1.0 FTE Community Relations and Assessment Manager who leads intake and community outreach. Each CSC site later integrated a 0.50 FTE Peer Mentor after securing foundation funding to cover peer role salaries [The peer role is further detailed in the discussion section below]. A Program Director, Senior Clinical Director, and Vice President also support teams. See Fig. 1 for

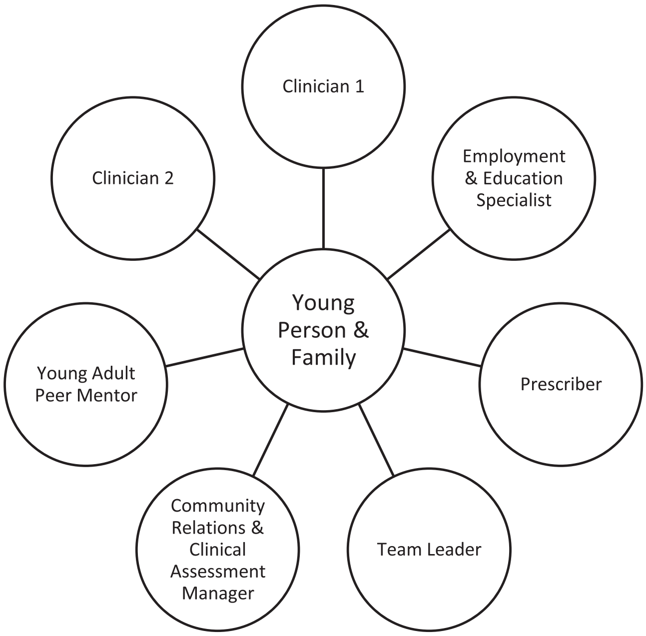

Fig. 1. MindStrong Team Roles.

current MindStrong team roles.

2. MindStrong service descriptions

2.1. Therapy and psychoeducation

Clinicians provided individual therapy and case management informed by Cognitive Behavior Therapy for psychosis (CBT-p) and Integrated Dual Disorder Treatment (IDDT). CBT-p is effective in first-episode psychosis (Hardy, 2017) and is prominent in the EDAPT and PIER CSC Models (NASMHPD, 2017). IDDT is a suggested best practice for treating first-episode psychosis and co-occurring substance use disorders (SAMHSA, 2019a). For individual and family psychoeducation, CSC staff used Individual Resiliency Training (IRT), a manualized NAVIGATE psychoeducational approach (Meyer et al., 2015). Clinicians hosted psychoeducation and support groups drawing from OnTrackNY (Bennett et al., 2018).

2.2. Vocational support

Vocational Specialists provided Young Adult Individual Placement and Support (IPS) and Supported Education (IPS Works, 2019). IPS has 8 principles, including zero exclusion (participation not based on symptom absence) and rapid job placement (face-to-face employer contact within one month). IPS and Supported Education are effective with the first-episode psychosis population (Humensky et al., 2019), and prominent in NAVIGATE and OnTrackNY CSC models (NASMHPD, 2017). One MindStrong team had a combined Employment and Education Specialist; the other had a part-time Employment Specialist and a part-time Education Specialist.

2.3. Psychiatry

MindStrong prescribers are either nurse-practitioners or psychiatrists trained in CSC prescribing best-practices ("start low, go slow"; Bello et al., 2017) who use shared-decision making practices (e.g., educating on benefits and side effects, co-developing an adherence plan, co-monitoring medication impact). These practices are foundational across CSC Models (Addington, 2021).

2.4. Program evaluation & quality improvement team

As part of the Thresholds Youth & Young Adult Services (YAYAS) (Thresholds YAYAS, 2021), MindStrong also has a dedicated evaluation team that: (1) used program evaluation and continuous quality improvement best-practices, (2) employed part-time young adults with lived and living mental health experiences to support program evaluation efforts, and (3) sponsored the YAYAS Advisory Board composed of 12 elected servicer participant representatives that advise on YAYAS policies, programs and practices akin to EASA's CSC Leadership Council (EASA, 2021).

3. Process for selecting key performance indicators & developing logic model

The YAYAS evaluation team partnered with YAYAS clinical leaders to develop the MindStrong logic model. See Fig. 2. The Thresholds YAYAS evaluation team reviewed the CSC academic literature, consulted with CSC experts, and leveraged the PhenX Toolkit, 2021 to bring operationalized definitions and measures of identified outcomes to YAYAS leaders. Together, clinical leaders and evaluation team picked Key Performance Measures (KPIs) to examine quarterly and annually. This led to an examination of service participation, and a request for analysis of service use and outcomes in their first year of enrollment. Analyses presented to clinical leaders and direct care staff on the first 12-months of MindStrong participation are presented in this paper.

This evaluation study adds to the very small body of research of CMHC efforts to implement CSC, examine participation and outcomes, and make adjustments. [There is only one evaluation study in Texas (North et al., 2019) detailing a university-affiliated CSC model design, implementation challenges, service participation type and frequency, and outcomes (n = 35).] This evaluation study aims to describe service participation and outcomes in the first year of CSC in a non-academic affiliated setting.

4. Methods

The University of Texas at Austin Institutional Review Board approved this project as a program evaluation for quality improvement. University researchers were not involved in the program evaluation design or implementation.

5. MindStrong CSC eligibility, recruitment & assessment

MindStrong adheres to FIRST.IL eligibility criteria. Participants must be between 14 and 40 years old and meet DSM-V criteria for a first psychotic episode in the last 18-months. Participants were ineligible if they had a primary neurodevelopmental disorder (e.g., autism), primary substance use disorder and/or medical condition (e.g., brain trauma). The Clinical Assessment Manager completed an in-person assessment to determine eligibility based on the Brief Psychiatric Rating Scale (BPRS; Ventura et al., 1993), which is frequently used in CSC research (PhenX Toolkit, 2021). After assessment, the Program Director, Psychiatrist and Clinical Assessment Manager met to determine eligibility.

6. Outcome measures and procedures

6.1. Service participation

MindStrong staff recorded service participation in the agency Electronic Health Care (EHR) record within seven days of service provision according to Illinois Medicaid requirements. Staff indicated service provision date, start and end time, and meeting details, including care plan goals addressed, staff interventions and participant responses.

Contextual Inputs	Core Strategies & Practices	Service Participation	Outcomes
• Agency experience implementing evidence- & team-based services • Agency experience blending & adapting models for youth • Embedded evaluation & quality improvement team • Medicaid & state block grant funding	• Blended evidence-based approaches • Cognitive Behavioral Therapy for Psychosis • Individual Resiliency Training • IPS Supported Employment & Education	• Total services • Clinician support services • Prescriber support services • Family support services • Vocational support services	• Increased developmentally appropriate education &/ employment engagement • Increased planned service exits • Avoid or decrease psychiatric hospitalization • Decreased symptom distress • Increased social & role functioning

Fig. 2. MindStrong Logic Model.

6.2. Service exits

Staff recorded participant exit types and related details in the agency EHR in real-time. Two research team members coded exits quarterly into the following categories (Klodnick et al., 2021a, 2021b): 1) Planned exits to: a) higher, b) similar, c) lower or d) no care levels and 2) Unplanned exits: a) to a similar level of care, b) premature and passive, c) after chronic missed meetings with self-termination or d) never engaged early on, or e) other. A third researcher reconciled coding differences.

6.3. Employment, education and psychiatric hospitalizations

At intake, staff recorded participant education employment, and psychiatric hospitalization histories in the EHR. Staff recorded education, employment, and psychiatric hospitalization start and stop dates, and related details, in the EHR in real-time.

6.4. Symptom and functioning measures

The State of Illinois requires FIRST.IL providers to complete the Clinician-Rated Dimensions of Psychosis Symptom Severity (CRDPSS) at enrollment and every six-months. The CRDPSS (Berendsen et al., 2020) has eight-items capturing psychotic symptom severity scored on a five-point scale (0 =none, 1 =equivocal; 2 =present, but mild; 3 =presents and moderate; and 4 =present and severe) and summed for a total score. Ratings reflect symptom severity over the past seven-days. Decreased scores indicate improvement. Clinicians also completed the Global Social and Role Functioning Scales (GFS and GFR; Cornblatt et al., 2007) within 30 days of enrollment and every six-months thereafter. The GFS and GFR assess social role functioning in interpersonal relationships and social contexts as well as role performance in work, school, and home (depending on age appropriateness) on two 10-point scales. Increased scores indicate improvement. Both scales have high construct validity and interrater reliability with those with a recent onset of psychosis (Piskulic et al., 2011), and are endorsed as core CSC research measures (Ongur et al., 2019).

7. Data analysis

7.1. Service participation

Service data for 47 participants' from their first 12-months of MindStrong enrollment were selected for analysis. This included: service codes, individual service dates and service length (in minutes). Across the 47 participants, there were 5206 services notes. Service types were coded into four categories: (1) clinical (e.g., assessment, case management, crisis intervention, psychoeducation, therapy); (2) family (e.g., psychoeducation, crisis intervention, therapy); (3) vocational (e.g., job/school exploration and support); and (4) psychiatry (e.g., medication monitoring). Descriptive statistics were used to summarize participation by month.

7.2. Service exits

The team summed each service exit code for those who exited services.

7.3. Employment, education and psychiatric hospitalizations

Education and employment engagement were coded as dichotomous variables (e.g., 0 =not employed; 1 =employed). Psychiatric hospitalizations were coded into a ranked variable (0 =no hospitalizations; 1 =1 hospitalization, 2 =2 hospitalizations; and 3 =3 or more hospitalizations).

7.4. Symptom and functioning measures

Standardized measure scores were analyzed, including: (a) 47 completed CRDPSS at baseline, 47 at six-months, and 41 at 12-months; (b) 47 participants completed GFS and GFR at baseline, 47 at six-months, and 43 at 12-months. Change in enrollment to 12-months scores were coded as "improved" or "not improved." Paired-sample t-tests were used to compare mean scores at baseline and 6-months, and baseline and 12-months.

8. Results

8.1. MindStrong participants

MindStrong enrolled 85 participants from 10/2016–3/2020. For this program evaluation, only those enrolled for at least 12-months ($n = 47$) were selected. Of the 47 enrolled for 12-months (Mean age=20.5, SD=3.8; Range=13–29 years). Two-thirds were Medicaid recipients, while one-third had commercial insurance, primarily through a parent. Prior to enrollment, 72% had at least one psychiatric hospitalization. See Table 1 for demographic information. Primary diagnoses included: Schizophrenia Spectrum (77%), Bipolar I (13%), Major Depressive (6%), and Posttraumatic Stress (4%) disorders. At enrollment, 47% were either in enrolled in school and/or employed (i.e., 30% were in school: eight in high school; six in post-secondary; 21% were employed: nine part-time;

Table 1
Participant Demographics.

Demographics (n = 47)	Count (%)
Gender	
Female	14 (30%)
Male	33, (70%)
Race/Ethnicity	
Black/African American	19 (40%)
White	16 (34%)
Latinx	5 (11%)
Asian	3 (6%)
Unknown	4 (9%)
Level of Education at Enrollment	
Bachelor's Degree	4 (8%)
Some College	12 (25%)
High School Diploma	15, 33%
Some High School	14, 29%
Unknown	2, 4%

one full-time).

9. MindStrong service participation

Individuals and families (n = 47) participated in, on average, 55 h of services (SD=23.5) in enrollment year 1. This was on average 4.70 h of services per month (SD=1.91). Monthly participation was higher on average for the first seven months of enrollment compared to months eight to 12. Fig. 3 and Table 2 summarize service participation by month.

10. Service participation by type

10.1. Clinician support services

In their first 12-months, 100% attended met at least once with their primary clinician. Participants engaged in, on average, 33.7 h of clinician-provided support (SD=14.5, Range=10.2–68.6) in their first year, which equates to 3.07 h per month on average (SD=0.51; Range=2.29–4.06). See Fig. 4.

10.2. Family support services

In their first 12-months, 57% of participants' families engaged in services at least once. For those who did (n = 27), it was for on average, 5.5 h (SD=5.0, Range=0.5–17.3) in their first year. Monthly family participation ranged from 2% (month 1) to 30% (in months 6 & 7). Family participation was greatest in quarter 2 (40%) compared to quarters 3 (32%), 4 (30%) and 1 (23%). On average, 9 families received support each month (SD=3.68; Range=1–14), for on average, 1.46 h per month (SD=0.29; Range=0.92–2.0). See Fig. 4.

10.3. Vocational support services

In their first 12-months, 68% engaged in employment and education services. For those who did (n = 33), it was for on average, 14.4 h (SD=11.9, Range=1.0–47.8). Approximately 15 participants (SD=4.64; Range=2–19) received vocational support each month, for on average, 2.49 h per month (SD=0.61; Range=1.75–3.62). See Fig. 4.

10.4. Medication support services

In their first 12-months, 87% participated in prescribing and medication monitoring services. For those who did (n = 41), it was for on average, 4.1 h (SD=2.6, Range=0.5–10.2). Approximately 21 participants (SD=2.67; Range=16–25) received medication services on average 0.68 h per month (SD=0.17; Range=0.49–1.10). See Fig. 4.

11. Service exits

Of the 85 enrolled in the selected time period (10/2016–3/2020), 38 exited before reaching 12-months of enrollment (approximate mean length of stay=6 months, SD=3 months). Most (63%) had planned exits: 42% to a lower care level; 16% to a similar care level; and 5% to higher care level (included supported housing). Approximately 37% had unplanned exits: 13% exited passively without termination, 13% had chronic missed meetings and self-terminated care, and 11% ceased participation after limited initial engagement.

12. Vocational engagement

In enrollment year 1, 66% were employed, 43% were enrolled in an education program, and 83% were employed or in school. Education programs included: post-secondary (n = 5), GED course (n = 3), and high school (n = 2). Twenty-five were employed part-time and six full-time. Employment was primarily entry level in the following areas: restaurant (e.g., serving, hosting, bussing), retail and grocery (e.g., cashier, stocking, customer service), delivery, receptionist and administrative support, mentoring and counseling, and information technology.

13. Psychiatric hospitalization

Most (72%) did not experience a psychiatric hospitalization in enrollment year 1. Of those who had a psychiatric hospitalization prior to enrollment (72%; 34 of 47), 23 (67%) were never hospitalized in their first year of enrollment; 10 (21%) had one psychiatric hospitalization, and one experienced four psychiatric hospitalizations.

14. Standardized measures

Scores on standardized measures all trended in desirable directions.

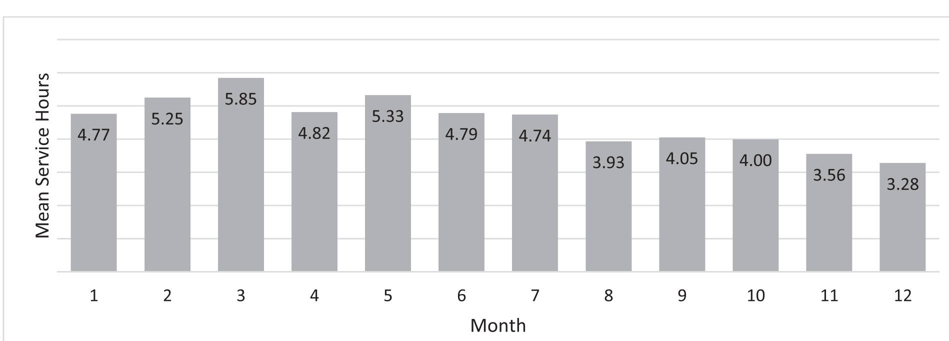

Fig. 3. Mean Hours of Service Participation by Month.

Table 2
Service Participation Details by Month.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Mean Hours	4.77	5.25	5.85	4.82	5.33	4.79	4.74	3.93	4.05	4.00	3.56	3.28
SD	3.61	3.73	3.53	2.90	3.59	2.74	3.03	2.93	3.00	3.23	2.66	2.37
Min	0.00	0.00	0.00	0.00	0.68	0.00	0.08	0.00	0.02	0.33	0.07	0.03
Max	16.62	18.38	17.30	14.55	19.78	11.8	15.02	12.43	13.18	14.17	16.18	10.78

*The zero's in the "min" row indicate that there were times when 1–2 participants did not participate in any services, but remained enrolled.

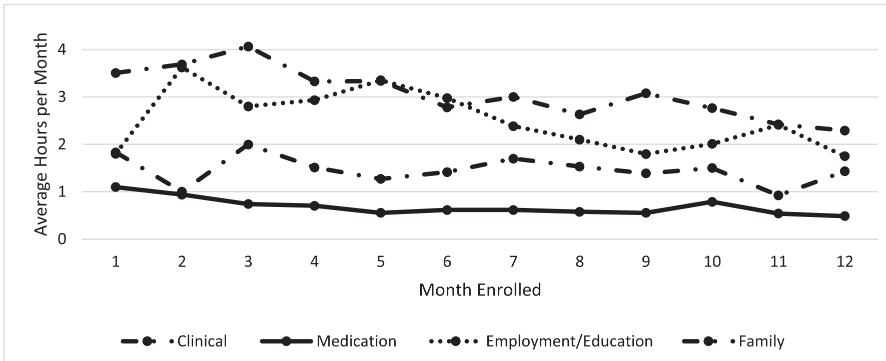

Fig. 4. Mean Service Hours by Type & by Month Enrolled.

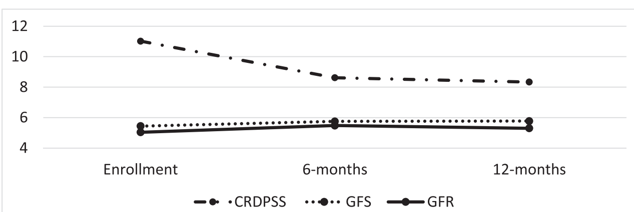

Fig. 5. Clinician-Rated Measure Mean Scores at Enrollment, Six- & 12-Months. [1]CRDPSS score decreases denote decreased symptom severity. [2]GFS and GFR score increases denote improved social and role functioning.

See Fig. 5 for standardized measure mean scores at baseline, six- and 12-months.

14.1. CRDPSS

From baseline to six-months ($n = 47$), 63% experienced decreased psychosis symptom severity as indicated by the CRDPSS; 13% had no change, and 24% experienced increased symptom severity. From baseline to 12-months ($n = 42$), 71% experienced decreased symptom severity as indicated by the CRDPSS; 5% had no change, and 24% experienced increased symptom severity. Paired sample t-tests revealed significant differences between: (1) CRDPSS baseline (10.93) and six-month (8.62) mean scores, $t(45) = 3.09$, $p = 0.003$; and 2) CRDPSS baseline (11.33) and 12-month (8.33) mean scores, $t(41) = 3.52$, $p = 0.001$.

14.2. Global Functioning Social (GFS)

As indicated by the GFS from baseline to six-months ($n = 47$), 57% increased in social functioning; 11% had no change, and 32% decreased in social functioning. No significant difference was observed between GFS baseline mean (5.44) and six-month mean (5.75), $t(46) = -1.95$, $p = 0.057$. As indicated by the GFS, from baseline to 12-months ($n = 42$), 55% increased in social functioning; 10% had no change, and 35% decreased in social functioning. No significant difference was observed between the GFS baseline mean (5.43) and 12-month mean (5.77), $t(41) = -2.04$, $p = 0.058$.

14.3. Global Functioning Role (GFR)

As indicated by the GFR, from baseline to six-months ($n = 46$): 63.0% increased in role functioning; 4.3% had no change, and 32.6% decreased in role functioning. No significant difference was observed between GFR baseline mean (5.04) and six-month mean (5.52), $t(45) = -1.94$, $p = 0.058$. As indicated by the GFR, from baseline to 12-months ($n = 41$): 65.9% increased in role functioning; 2.4% had no change, and 31.7% decreased in role functioning. No significant difference was observed between the GFR baseline mean (4.91) and 12-month mean (5.33), $t(40) = -1.32$, $p = 0.194$.

15. Discussion

This study highlights metrics that mattered to CSC site leaders in a non-university affiliated and fee-for-service billing community mental health care center. Importantly, rates of observed CSC service utilization, exits, psychiatric hospitalization and employment and education were similar to those found in U.S. research and state-wide CSC demonstration projects. CSC site leader engagement with the data summaries presented in this paper (and similar quarterly KPI summary reports), led to several program enhancements that warrant sharing as well as further evaluation. Study findings also have implications for NIH EPINET, a national university-driven data harmonization and healthcare learning network (EPINET, 2021), and non-academic affiliated community mental health centers (CMHCs) interested in demonstrating positive outcomes in order to successfully sustain CSC.

Program evaluations like these contribute to the field's limited understanding of CSC participation by type and frequency over-time. Participant engagement in approximately 4.7 service hours per month, and accessing higher service rates in the first six months, seems reasonable following psychosis onset. Observed service rates were similar to those observed in a Texas CSC demonstration project (Mean=48.34 h/year; $n = 35$) when "case management" services (Mean=40.40 h/year) were removed (North et al., 2019). MindStrong did provide some services in the community, but did not provide intensive case management or community support akin to the Texas site.

MindStrong has since integrated part-time Peer Mentors to: provide non-stigmatizing avenues for skill development and community-based support while fostering clinical and vocational service engagement. MindStrong Peer Mentors use a combination of OnTrackNY's Peer Support Model (DuBrul et al., 2017) and Massachusetts Peer Mentor Model (Young Adult Peer Mentor Practice Profile Workgroup, 2017). Peer Support is increasingly integrated into CSC across the country (SAMHSA, 2019b), and further understanding how integrating peers influences service participation and outcomes is warranted. Also, during the COVID-19 pandemic, video-conferencing became a primary MindStrong service delivery mode. MindStrong leadership is currently wrestling with how to strategically blend video-conferencing with in-person support to better meet participant's mental health and developmental needs – and to maximize service engagement.

Observed CSC vocational service participation (68%) and vocational engagement (83%) in this evaluation are similar to rates found in NIH RAISE CSC studies. For example, among OnTrackNY participants (n = 634), 82% engaged in vocational services and 75% were enrolled in school and/or employed in their first year (Humensky et al., 2019). And, 68% of (n = 220) NAVIGATE participants participated in three or more vocational services in their first six-months and 43% were in school or employed (Rosencheck et al., 2017). Akin to OnTrackNY CSC findings (Humensky et al., 2019), participants in this evaluation engaged in vocational services earlier in their enrollment. However, this study detailed vocational service hours on average across the year (approximately 14 h) and by month (approximately 2.5 h) for just over half the sample who engaged with a vocational specialist. This is helpful for better understanding CSC care experiences, in particular for sites that aim to provide high fidelity IPS Supported Employment and Education (see IPS Employment Center, 2019 for fidelity scale). MindStrong Vocational Specialists were supervised by IPS Team leaders within Thresholds Employment Department to ensure high fidelity. Both MindStrong sites have since combined Supported Employment-Supported Education roles, and increased role to full-time, to be more in-line with national best-practice and effectively support career development.

Family participation (57%) in at least one psychoeducation or support session appeared less than previous studies, including: 83% in the Washington State Journeys CSC evaluation (Oluwoye et al., 2019) and 94% in OnTrackNY CSC study (Drapalski et al., 2017). More consistent monthly family participation in services was also lower (19%) compared to 25–45% of families in NAVIGATE CSC (Jones et al., 2018). Although MindStrong did offer group family psychoeducation at times, it was not consistent due to CSC staff and leadership turnover. (The "family education and support" variable in this evaluation combined individual and group family support). Since this evaluation, CSC site staff were trained in the nuances between family support and family therapy, and how to assertively engage families in CSC services. The MindStrong Team Leader provides family therapy, and the opportunity for newer clinicians to observe their sessions to increase new clinician family engagement skills. And, a clinician at each site has been identified as a "family champion." This clinician engages in available family-focused CSC trainings and consults with teammates to improve their family engagement strategies.

The 37% unplanned MindStrong exit rate was comparable to CSC research trials and state evaluations - where it is estimated between 12% and 52% disengage prematurely (Mascayano et al., 2021). There is an immense need to better understand how and why CSC exits occur. Research suggests that passive/unplanned exits are related to a number of individual, family, and program-related factors (Jones et al., 2020). Anecdotally, Thresholds observed when CSC participants make progress (e.g., living more independently, returning to school and/or work, finding a medication that works), participants tend to decrease participation and/or do not see a need for services. Improved conceptualization of CSC treatment engagement over-time and examination of step-down CSC programming are important for preventing unplanned exits. Easy to access step-down programming is ideal, but difficult to fund using static fee-for-service billing models. Thresholds MindStrong leaders identified "long-stayers" as those who were enrolled for over 2-years, and monitored their work and school engagement, hospitalizations, and measure scores to determine if they should "step-down" in care. MindStrong leaders are also currently discussing how to alternate video-conferencing with in-person support as a potential step-down practice.

The standard for CSC provision frequency is at least a once a week connection between team and client (Addington, 2021). In theory, this could be with any of the multidisciplinary team members based on individual (and family) needs. For CSC programs like MindStrong that fiscally rely heavily on Medicaid and commercial insurance billing (and less on grant funding) to operate, there is a necessary focus on promoting CSC licensed clinician and prescriber engagement. These staff bill at higher rates than education and employment specialists, case managers and peer providers. The result is an increased emphasis on symptom reduction, therapy participation, and medication adherence – which may inadvertently decrease engagement. Young people seek care for other social determinants of health. For example, treatment goals among young adults enrolled in a sister team to MindStrong (called Emerge) included: finding jobs and enrolling in school, living more independently, and improving relationships (Klodnick et al., 2021a, 2021b). To that end, Thresholds began training MindStrong staff in the Transition to Independence (TIP) Model (Dresser et al., 2015) to ensure that staff approached their work first from a youth and young adult developmental lens. And, the Peer Mentor integration into MindStrong may provide for alternative ways of making sense of and understanding mental health experiences, wellness and healing, that ultimately may promote engagement.

15.1. Limitations & next steps

This program evaluation was limited to two CSC sites located in urban and suburban areas, and only included year 1 CSC engagement and outcomes. The CSC sites often had unreliable historical data regarding duration of untreated psychosis, which is why it is not included here. Most of the sample did not have the opportunity to participate in peer support (it was added in 2019), which is why it is not included here. Although the two CSC sites appeared in strong alignment with CSC fidelity, this program evaluation did not employ a fidelity scale (e.g., Addington et al., 2021). Despite low rates of psychiatric hospitalizations and decreases in symptom severity, minimal changes in GFS and GFR over time were observed. This may be due to clinician turnover and lack of inter-rater reliability meetings. Thresholds partnered with a university-affiliated consultant to provide measure administration and scoring training; but to date, has not instituted regular inter-rater reliability meetings due to the many training and administrative-related tasks required of staff for fiscally sustaining services. And, finally a major limitation is that we did not examine the relationship between KPIs and service type, amount, or mix overtime. This is particularly challenging given that MindStrong services are dynamic and tailored to individual and family needs and availability (i.e., some may want or need less or more support depending on prioritized needs and treatment goals), as well as, historically high turnover rates among community mental health direct practice staff. Our next step as an evaluation team is to match service data with workforce data, and with expert statistician support, to investigate service participation patterns and associated KPI changes.

15.2. Implications for CSC sustainability

To provide CSC, CMHCs currently braid federal, state, county, foundation, donor, Medicaid and commercial insurance funding (Bao et al., 2021). Providers must capture and report on outcomes for funders. There are many barriers to systematically collecting, analyzing and using data in CMHC settings. Providers often lack the technology (e.g., nimble electronic health care record), workforce resources (e.g., team supervisor's time) to manage systematic standardized measure completion, and the expertise and oversight to ensure measure validity and reliability (Cohen, 2011, 2015). Staff in fee-for-service settings tend to have maximized caseloads and less time for additional administrative tasks. CMHCs face high turnover rates due to inadequate pay and benefits, and stressful working conditions – which inevitably impact CSC data collection.

This program evaluation took place at Thresholds, a CMHC with a long history of university partnership. Thresholds has worked to effectively address many of the above listed data collection barriers. Thresholds employs a PhD-level researcher who partners with quality, information technology, and clinical departments to: 1) design and maintain data systems, 2) translate data in useful reports, and 3) support program report use. With strong financial and workforce investment in cross-departmental collaboration and program evaluation, CSC evaluation appears feasible in non-academic settings. Illinois recently passed legislation to establish Medicaid-based and commercial insurance payment mechanisms to improve adolescent and young adult team-based service flexibility and funding (Illinois General Assembly, 2018). Earmarked financial resources for CMHC-based CSC evaluation must be included in CSC financial models.

16. Evaluation lessons learned

This study produced a number of lessons learned for successfully evaluating CSC in non-academic settings. First, is the critical importance of CSC program managers and team leaders in supporting data collection as "practice as usual" in program operations, and not perceiving life event documentation (e.g., psychiatric hospitalization start and end dates) or standardized measure (e.g., CRDPSS) completion as "extra." Historically, program leaders collected data for grants (e.g., federal, state, foundation) and had not seen efforts result in useful data summaries. CSC sites that aim to integrate additional measures for program evaluation purposes must embed CSC KPI tracking with other key document tracking (e.g., annual consent for services; bi-annual care plan updates). Second, is the importance of regular program leader sponsored data validation, cleaning and on-going monitoring to ensure data accuracy. On multidisciplinary teams, multiple staff are responsible for data input, and there is great opportunity for data entry to be missed, in particularly around documenting real-time life events (e.g., psychiatric hospitalizations). CSC sites must establish practices for which staff on the team enters participant life events in the EHR. Thresholds also instituted a monthly data validation check where participant KPIs were reviewed and corrected by CSC site leaders. Third, depending on an agency's EHR flexibility and investment in tailoring the EHR to small programs like CSC, measures and life events may need to be collected and tracked outside of the EHR. This is achievable using fillable PDFs, a HIPPA compliant version of SurveyMonkey, creating simple trackers in EXCEL and training staff to use, and designing systems for effective program leader oversight. And finally, simple, easy to interpret, individual participant, site, and cross-site data summaries are critical in order to foster data use to improve individual clinical care and program functioning. This requires a deep partnership, and an inclusive design and refinement process, between clinical leaders, direct care staff, and program evaluation team.

This study has particular implications for a large national effort to better understand and improve CSC. Led by university researchers and funded by NIH, EPINET (2021) aims to harmonize data collection and generate a national early psychosis learning network. EPINET has eight university-located hubs that partner with 101 CSC sites to collect the Core Assessment Battery (CAB), which includes multiple clinician- and participant-completed standardized assessments (EPINET, 2021). EPINET is recruiting CMHCs to participate in CAB collection. National data collection efforts that seek to effectively include CMHCs must provide financial resources, technical assistance, and training to: 1) aid providers in developing systems, electronic and informatics platforms to collect measures in valid and reliable ways, and 2) prevent additional staff burden and turnover in these already fiscally-strained and stressful work settings.

17. Conclusion

CSC program evaluation is possible in well-resourced community settings. Many engaged in CSC, made vocational progress and experienced decreased symptoms and hospitalizations. Future research examining the relationship between service participation and outcomes is needed. Financial resources are needed (beyond service cost) to capture (and examine) CSC engagement and outcomes in CMHCs. CSC is promising for reducing long-term adult service system utilization nationally. Investment in improving CSC evaluation and sustainability in CMHCs is vitally important.

Ethics board approval statement

The University of Texas at Austin Institutional Review Board approved this program evaluation, which analyzed de-identified agency electronic healthcare records.

CRediT authorship contribution statement

Vanessa V. Klodnick: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Software, Writing – original draft. **Rebecca P. Johnson:** Conceptualization, Investigation, Methodology, Writing – original draft. **Ariel Brenits**: Data curation, Formal analysis, Visualization, Writing – review & editing. **Margaret Ann Pauuw**: Data curation, Formal analysis, Visualization, Writing – original draft. **Vanessa V. Kodnick**: Funding acquisition. **Marc A. Fagan:** Funding acquisition, Resources, Software, Supervision, Writing – review & editing. **Eva Zeidner**: Resources, Validation, Writing – review & editing. **Deborah A. Cohen**: Supervision, Validation, Writing – review & editing.

Acknowledgements

The authors thank: (1) former Thresholds YAYAS Research & Innovation intern, Ethan Husbands, for data cleaning support, and (2) current and previous Thresholds MindStrong leaders for their contribution to building, evaluating, and refining MindStrong: Candy Malina, Kelly Alexander, Brittany Tolar, and Jose Viruet.

Declarations

Partial financial support was received from the State of Illinois Mental Health Block Grant Set-aside for Early Intervention in Serious Mental Illness awarded to Thresholds to provide early psychosis care and collect measures and key performance indicators.

Responsibility statement

All authors certify responsibility for this manuscript's content and writing.

Financial interests

The authors declare they have no financial interests.

Non-financial interests

none.

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Eva Zeidner, LCPC. Eva has successfully used her marketing background to effectively design mental health education and service orientation information for young people with serious mental health needs, as well as her 20 + years in community mental health to refine Thresholds Youth & Young Adult Services (YAYAS) Emerging Adult screening, assessment and intake processes. Eva currently serves as the YAYAS Community Relations & Clinical Assessment Manager, and previously held the YAYAS Emerging Adults Programs Assistant Director for 3.5 years. Eva completed her masters in counseling psychology at Northwestern University and is licensed in Illinois.

Marc A. Fagan, PsyD. Dr. Fagan served as the Vice President of Clinical Operations for Thresholds Youth & Young Adult Services for 18 years where he built nationally recognized multidisciplinary service models for 16-26 year olds diagnosed with serious mental health conditions that blend and adapt child and adult evidence-based practices. Dr. Fagan participates in multiple U.S. national and state workgroups dedicated to improving transition-age youth mental healthcare policies and practices. Dr. Fagan has trained providers in over 10 states. Dr. Fagan is now the Plan President of YouthCare at Centene. Dr. Fagan completed his doctorate at the Illinois School of Professional Psychology.